MEDIUM · ADVANCED MATH · ANSWER KEY

# Equivalent expressions

# 02

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ANSWER KEY · SAT QUESTION BANK

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# Q1

A

A. $4x^2+21x+33$

Choice A is correct. The given expression can be rewritten as $(2x+5)^2+(-1)(x-2)+2(x+3)$. Applying the distributive property, the expression $(-1)(x-2)+2(x+3)$ can be rewritten as $-1(x)+(-1)(-2)+2(x)+2(3)$, or $-x+2+2x+6$. Adding like terms yields $x+8$. Substituting $x+8$ for $(-1)(x-2)+2(x+3)$ in the given expression yields $(2x+5)^2+x+8$. By the rules of exponents, the expression $(2x+5)^2$ is equivalent to $(2x+5)(2x+5)$. Applying the distributive property, this expression can be rewritten as $2x(2x)+2x(5)+5(2x)+5(5)$, or $4x^2+10x+10x+25$. Adding like terms gives $4x^2+20x+25$. Substituting $4x^2+20x+25$ for $(2x+5)^2$ in the rewritten expression yields $4x^2+20x+25+x+8$, and adding like terms yields $4x^2+21x+33$.

Choices B, C, and D are incorrect. Choices C and D may result from rewriting the expression $(2x+5)^2$ as $4x^2+25$, instead of as $4x^2+20x+25$. Choices B and C may result from rewriting the expression $-(x-2)$ as $-x-2$, instead of $-x+2$.

## Q2 B

B. $\sqrt[144]{a^{132}}$

Choice B is correct. Since $\frac{12}{12} = 1$, multiplying the exponent of the given expression by $\frac{12}{12}$ yields an equivalent expression: $a^{\frac{11}{12}\frac{12}{12}} = a^{\frac{132}{144}}$. Since $\frac{132}{144} = 132\frac{1}{144}$, the expression $a^{\frac{132}{144}}$ can be rewritten as $a^{132\frac{1}{144}}$. Applying properties of exponents, this expression can be rewritten as $a^{132\frac{1}{144}}$. An expression of the form $m^{\frac{1}{k}}$, where $m > 0$ and $k > 0$, is equivalent to $\sqrt[k]{m}$ . Therefore, $a^{132\frac{1}{144}}$ is equivalent to $\sqrt[144]{a^{132}}$ .

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

## Q3 D

D. $4x^{-1}y^4$

Choice D is correct. Taking the square root of an exponential expression halves the exponent, so $\frac{\sqrt{16x^4y^8}}{x^3} = \frac{4x^2y^4}{x^3}$, which further reduces to $\frac{4y^4}{x}$. This can be rewritten as $4x^{-1}y^4$.

Choice A is incorrect and may result from neglecting the denominator of the given expression and from incorrectly calculating the square root of 16. Choice B is incorrect and may result from rewriting $\frac{1}{x}$ as $x^1$ rather than $x^{-1}$. Choice C is incorrect and may result from taking the square root of the variables in the numerator twice instead of once.

## Q4

B

**B.** $x^4 + 23x^2 + 96$

Choice B is correct. The expression $x^2 + 11^2$ can be written as $x^2 + 11x^2 + 11$, which is equivalent to $x^2x^2 + 11 + 11x^2 + 11$. Distributing $x^2$ and 11 to $x^2 + 11$ yields $x^4 + 11x^2 + 11x^2 + 121$, or $x^4 + 22x^2 + 121$. The expression $x - 5x + 5$ is equivalent to $x - 5x + x - 55$. Distributing $x$ and 5 to $x - 5$ yields $x^2 - 5x + 5x - 25$, or $x^2 - 25$. Therefore, the expression $x^2 + 11^2 + x - 5x + 5$ is equivalent to $x^4 + 22x^2 + 121 + x^2 - 25$, or $x^4 + 22x^2 + 121 + x^2 - 25$. Combining like terms in this expression yields $x^4 + 23x^2 + 96$.

Choice A is incorrect. Equivalent expressions must be equivalent for any value of $x$. Substituting 0 for $x$ in this expression yields -14, whereas substituting 0 for $x$ in the given expression yields 96.

Choice C is incorrect. Equivalent expressions must be equivalent for any value of $x$. Substituting 0 for $x$ in this expression yields 121, whereas substituting 0 for $x$ in the given expression yields 96.

Choice D is incorrect. Equivalent expressions must be equivalent for any value of $x$. Substituting 0 for $x$ in this expression yields 146, whereas substituting 0 for $x$ in the given expression yields 96.

## Q5

B

**B.** $xy^{\frac{9}{7}}$

Choice B is correct. For positive values of $a$ and $b$, $a^m b^m = ab^m$, $\sqrt[n]{a} = a^{\frac{1}{n}}$, and $a^{j^k} = a^{jk}$. Therefore, the given expression, $\sqrt[7]{x^9y^9}$, can be rewritten as $\sqrt[7]{xy^9}$. This expression is equivalent to $xy^{9^{\frac{1}{7}}}$, which can be rewritten as $xy^{9 \cdot \frac{1}{7}}$, or $xy^{\frac{9}{7}}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

## Q6

B

B. $\frac{8x - 3}{2}$

Choice B is correct. The given expression has a common factor of 2 in the denominator, so the expression can be rewritten as $\frac{8xx - 7 - 3x - 7}{2x - 7}$. The three terms in this expression have a common factor of $x - 7$. Since it's given that $x > 7$, $x$ can't be equal to 7, which means $x - 7$ can't be equal to 0. Therefore, each term in the expression, $\frac{8xx - 7 - 3x - 7}{2x - 7}$, can be divided by $x - 7$, which gives $\frac{8x - 3}{2}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

## Q7

A

A. $x^3 + 10x$

Choice A is correct. Applying the distributive property, the given expression can be written as $7x^3 + 7x - 6x^3 + 3x$. Grouping like terms in this expression yields $7x^3 - 6x^3 + 7x + 3x$. Combining like terms in this expression yields $x^3 + 10x$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

## Q8

C

**C.** $8d^3 - 48d^2 - 3d + 18$

Choice C is correct. Applying the distributive property to the given expression yields $d8d^2 - 3 - 68d^2 - 3$. Applying the distributive property once again to this expression yields $d8d^2 + d\text{-}3 + \text{-}68d^2 + \text{-}6\text{-}3$, or $8d^3 + \text{-}3d + \text{-}48d^2 + 18$. This expression can be rewritten as $8d^3 - 48d^2 - 3d + 18$. Thus, $d - 68d^2 - 3$ is equivalent to $8d^3 - 48d^2 - 3d + 18$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

## Q9

A

**A.** 7

Choice A is correct. Since the given expressions are equivalent and the numerator of the second expression is $\frac{1}{6}$ of the numerator of the first expression, the denominator of the second expression must also be $\frac{1}{6}$ of the denominator of the first expression. By the distributive property, $\frac{1}{6}6x + 42$ is equivalent to $\frac{1}{6}6x + \frac{1}{6}42$, or $x + 7$. Therefore, the value of $b$ is 7.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

## Q10 C

C. $x+10$

Choice C is correct. Distributing the negative sign to the terms in the second parentheses yields $(2x+3)-x+7$. This expression can be rewritten as $2x-x+3+7$. Combining like terms results in $x+10$.

Choice A is incorrect and may result from not distributing the negative sign to the 7. Choice B is incorrect and may result from adding $(x-7)$ to $2x+3$ instead of subtracting $(x-7)$. Choice D is incorrect and may result from adding the product of $2x$ and x to the product of 3 and 7.

### Notes

Accept equivalent forms (e.g. `0.25` ≡ 1/4). For grid-in items, decimal and fraction answers are both valid.

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